Bipolar measurement of affective valence does not differentiate neutral and ambivalent responses

Department of Psychology[1], Department of Brain Cognitive Science[2]
Taeyang Lee[1], Jinwoo Lee[1], Minseok Koo[2], Jiook Cha[1,2], Sangah Lee[2]
rlaxodid5539@snu.ac.kr

INTRODUCTION

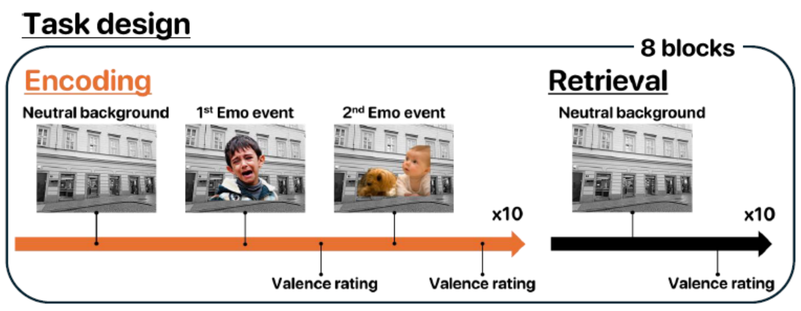
- There are numerous reports suggesting that **ambivalent affect has distinct neural representations** and physiological response profiles at the cortical level, compared to simple neutral, positive, or negative affect(Lettieri et al., 2019; .Vaccaro et al., 2024).
- However, the commonly used **bipolar measurement of affective valence does not allow for ambivalent responses** at the self-report level.
- As a result, it implicitly encourages participants to choose which valence is more dominant in ambivalent states, leading to a dichotomous evaluation that fails to clearly capture such ambivalent experiences(Schneider et al., 2016).
- Therefore, we designed the stimuli that combine conflicting valence information and hypothesized that **such ambivalent responses would be distinguishable from neutral and univalent responses at the neural level**. However, at the self-report level, using bipolar measurements, they would not be differentiated from neutral responses.

METHODS

- **Data Collection**

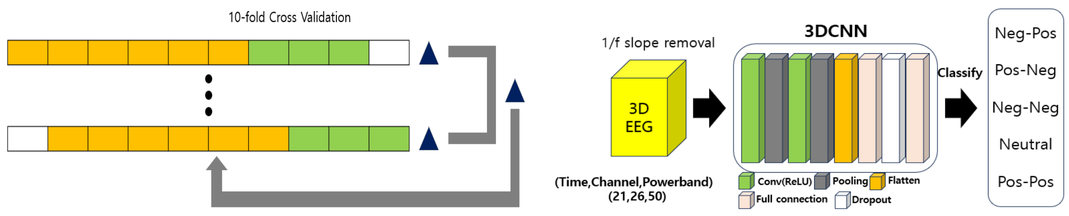
Participants : 57 (31 Female), Age : 25.35 (*SD* = 4.23)

Collect 32 Channels EEG data and Use 26 Channel's data for analysis



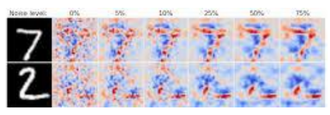
- **Method of Analysis**

1. Training an emotional stimuli classifier based on 3D CNN.



Since brain responses related to emotions exhibit nonlinear characteristics, we opted for a nonlinear approach by training the classifier using deep learning(Lin et al., 2023; Viinikainen et al., 2010).

2. Model analysis through Integrated Gradients.

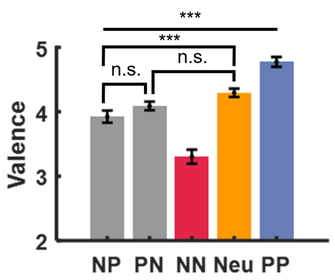


By this process, we can identify which features are important for emotion classification.

(Smilkov et al., 2017)

RESULTS

- **Behavioral Results**

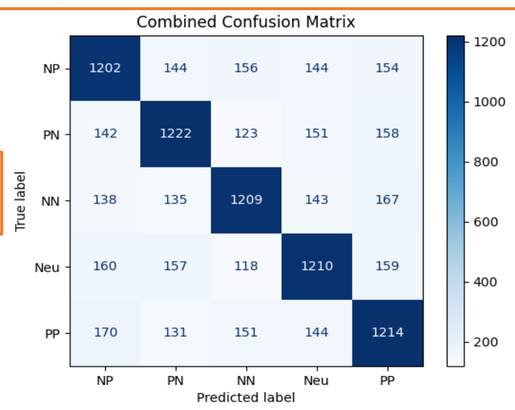


- Neg-Pos & Pos-Neg
  Mean difference = -0.159, t = -1.870 (p<.628)
- Pos-Neg & Neutral
  Mean difference = -0.197, t = -2.305 (p<.221)
- Neg-Pos & Neutral
  Mean difference = -0.356, t = -4.175 (p<.001)

The overall pattern was 'NN < NP < PN < Neutral < PP,' with no significant differences observed between Neg-Pos and Pos-Neg, and between Pos-Neg and Neutral.

- **Model Training Results**

1) Confusion matrix

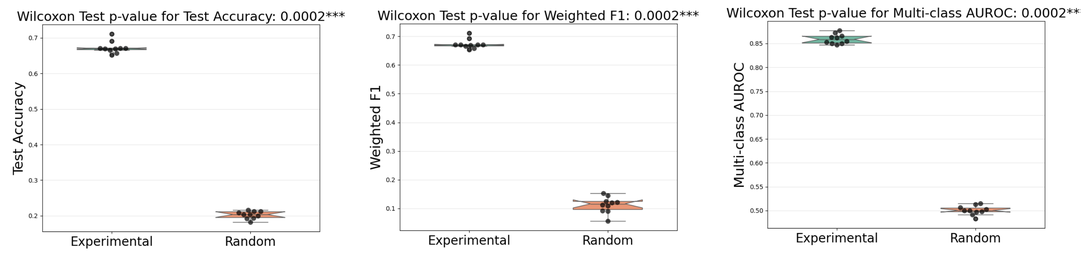


The model demonstrated balanced performance across all emotional conditions.

2) Results of Model's Test Performances

Metric	Test Accuracy	Weighted F1	Multi-class AUROC
Trained model	**0.6729** (*SD*=0.0160)	**0.6731** (*SD*=0.0161)	**0.8596** (*SD*=0.0096)
Random model	0.2024 (*SD*=0.0101)	0.1126 (*SD*=0.0268)	0.5007 (*SD*=0.0091)

3) Results of Wilcoxon rank sum test



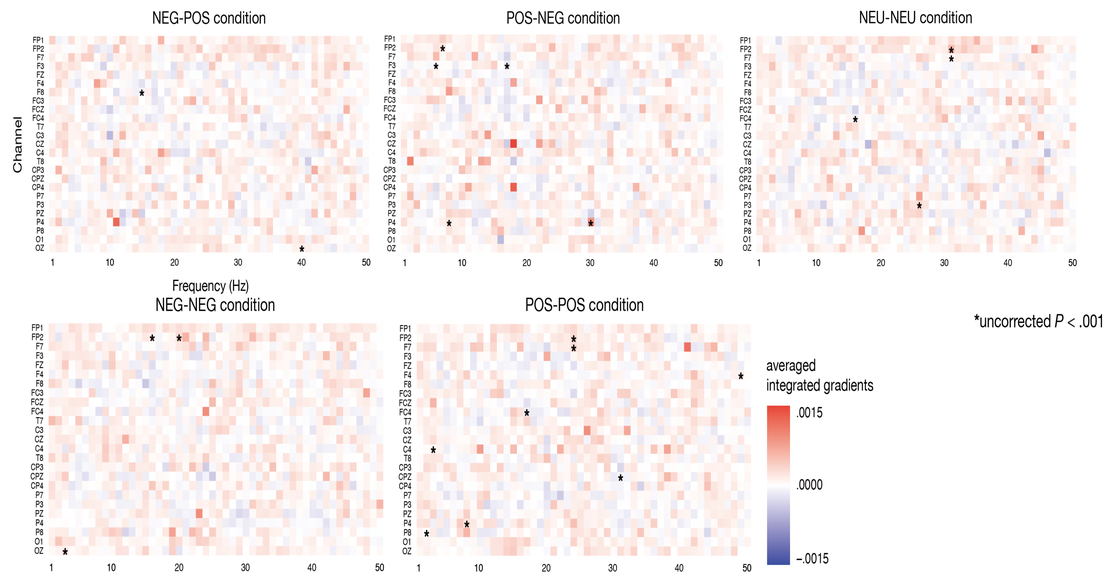
- **Model Analysis Results**

1) Variance contribution

Time : 0.0149, Channels : 0.0421, Powerband : 0.1634

Pick dimensions of channels and powerband to make 2D map

2) Difference of Attribution Maps



The channel-frequency maps representing emotional conditions differed across all emotional conditions. The neural responses for Neg-Pos, Pos-Neg, Neg-Neg, Neutral, and Pos-Pos conditions are all distinct.

DISCUSSIONS

- Ambivalent emotions exhibited a more negative valence compared to neutral emotions. In particular, the **Neg-Pos condition showed lower valence than neutral**, which can be explained in the context of **Negativity Bias**(Norris, 2021; Waugh et al., 2019).
- EEG patterns differed distinctly across the emotional conditions.

1) **The partial activation of frontal beta waves observed in the Neg-Neg condition** especially in right hemisphere is associated with negative emotions(Güntekin, B., & Başar, E., 2010).

2) **The high beta wave patterns observed in the Pos-Pos condition** are associated with positive emotions(Mishra et al., 2022).

3) **The part of frontal theta waves observed in the Pos-Neg condition** is associated with emotional regulation, suggesting a connection to ambivalent emotion(Park et al., 2011).

- This study highlights **the limitation of using a simple 1D valence measure in accurately linking the neural differences** observed between neutral and ambivalent emotions to behavioral outcomes.
- Future research could explore whether nonbipolar measurements, such as a 2D valence scale, can better capture these subtle differences.

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